

Accessing Your Applications from Outside

Ingress & Egress — Module 6 of 13

Enterprise EKS Platform





Module Agenda

- Why External Access Matters
- NGINX Ingress Controller & IngressClass
- Ingress Resources: Path-Based & Host-Based Routing
- Annotations, TLS Termination & cert-manager
- Gateway API: GatewayClass, Gateway, HTTPRoute
- Traffic Splitting and Canary Deployments
- Enterprise Integration & Egress Controls



Why External Access Matters

Getting Traffic Into Your Cluster

External Access Options

Method	Layer	Use Case	Limitation
NodePort	L4	Dev/testing	Port range 30000-32767, exposed on all nodes
LoadBalancer	L4	Single service exposure	One LB per service (\$\$\$), no path routing
Ingress	L7	HTTP/HTTPS routing	HTTP-only, limited extensibility
Gateway API	L4-L7	Advanced routing	Newer, still maturing ecosystem



NGINX Ingress Controller

Architecture and IngressClass

NGINX Ingress Architecture

How It Works

1. NGINX Ingress Controller pod runs NGINX reverse proxy
2. Exposed via a LoadBalancer Service (single AWS NLB/ALB)
3. Watches for Ingress resources across the cluster
4. Dynamically generates NGINX configuration from Ingress rules
5. Reloads NGINX when Ingress resources change

IngressClass Configuration

What IngressClass Does: Decouples the Ingress resource from the controller. Setting `is-default-class` means any Ingress without an explicit `ingressClassName` is handled by that controller.

Recommended Pattern: Run separate IngressClasses for external (internet-facing) and internal (VPC-only) traffic, each with its own controller and load balancer.



Ingress Resources

Path-Based and Host-Based Routing

Path Types Explained

pathType	Matching Behavior	Example
Exact	Matches URL path exactly	/api matches /api only, not /api/v1
Prefix	Matches path prefix by / segments	/api matches /api, /api/v1, /api/v1/users
ImplementationSpecific	Depends on IngressClass controller	NGINX supports regex: /api(/ \$)(.*)

Ingress Annotations

Customizing NGINX Behavior

TLS Termination

Securing Ingress with HTTPS

Gateway API

The Evolution of Kubernetes Ingress

Ingress vs Gateway API

Aspect	Ingress	Gateway API
Extensibility	Annotations (untyped strings)	Typed, structured API fields
Role model	Single resource, one owner	Role-oriented (infra, cluster, app)
Protocols	HTTP/HTTPS only	HTTP, TCP, UDP, gRPC, TLS
Traffic splitting	Not native	Built-in weight-based routing
Header matching	Not supported	Native header/query matching
Portability	Vendor-specific annotations	Portable across implementations
Status	Stable, feature-frozen	GA (HTTPRoute), evolving



Gateway API Resource Model



GatewayClass

Role: Infrastructure
Provider

- Defines controller implementation (cluster-scoped)



Gateway

Role: Cluster Operator

- Configures listeners, TLS, and provisions LB



HTTPRoute

Role: Application
Developer

- Defines routing rules; attaches to Gateway

Role Separation: Infrastructure teams manage GatewayClass and Gateway. Application teams create HTTPRoutes. RBAC applies cleanly at each level.



Traffic Splitting

Canary Deployments with HTTPRoute Weights



Enterprise Integration

Envoy Gateway + External DNS + cert-manager



Enterprise EKS Ingress Stack



Envoy Gateway

- Gateway API + Envoy data plane
- Advanced L7: gRPC + HTTP



External DNS

- Watches Gateway/Ingress
- Auto-creates Route53 records



cert-manager

- Automated cert lifecycle
- Internal CA, ACME, or Gateway API

Workflow: HTTPRoute → Envoy configures proxy → External DNS creates Route53 record → cert-manager provisions TLS → App is live.

Egress Controls

Managing Outbound Traffic



Why Egress Controls Matter



Security Concerns

- Compromised pods phoning home or exfiltrating data
- Supply chain attacks reaching external registries
- Compliance requirements (PCI-DSS, SOC2)



Control Mechanisms

- **NetworkPolicy egress rules** (Module 8)
- **Egress gateways** (service mesh) or **NAT gateways**
- **Proxy servers** for HTTP traffic

Egress Patterns on EKS

NAT Gateway (Default EKS)

- Pods egress via the VPC NAT gateway with a fixed Elastic IP
- All egress shares the NAT GW IP — useful for IP allowlisting
- No application-level control

Custom CA for External Endpoints

- Mount a CA bundle Secret into pods for mTLS with external APIs
- Trust private CAs without rebuilding container images



Lab 6: Ingress and Gateway API

Hands-On Exercise

- Verify the NGINX Ingress Controller and IngressClass
- Create host-based and path-based Ingress resources
- Configure TLS termination with a self-signed certificate
- Test Ingress annotations: rate limiting, CORS, rewrite
- *Optional:* Deploy Gateway API resources with HTTPRoute traffic splitting
- *Optional:* Configure egress NetworkPolicies to control outbound traffic

Duration: ~30-40 minutes

Key Takeaways

1. Consolidate with Ingress/Gateway — Avoid one LoadBalancer per service. Route through a shared entry point.

2. Gateway API is the future — Typed resources, role separation, and native traffic management replace annotation-driven Ingress.

3. Automate TLS — Use cert-manager for certificate lifecycle. Avoid managing certificates manually.

4. Control egress too — Ingress is only half the story. Egress controls are critical for security and compliance.

? Questions & Discussion

Module 6: Accessing Your Applications from Outside — Ingress
& Egress

Up Next: Module 7 — RBAC & Security